Performance Testing

Date	5 JULY 2026
Team ID	
Project Name	Smart lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	User Login, Loan Application, Fraud Detection API
Test Environment	Local Server (Flask + MySQL)
Test Date	5 july 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	50	60	Successful login with response time under 2 seconds

2	Loan Application Submission	100	120	Applications processed successfully without errors
3	Fraud Detection API	50	90	Fraud score generated correctly for each request
4	Loan Approval Process	30	60	Loan approval completed successfully with stable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	pass	Meets performance target
2	Response Time (Max)	< 5 seconds	3.8 sec	pass	Within acceptable limit
3	Throughput (Req/sec)	100 Req/sec	118 Req/sec	pass	Good throughput
4	Error Rate	< 1%	0.4	pass	Very low error rate
5	CPU Utilization	< 80%	68	pass	Stable CPU usage
6	Memory Utilization	< 80%	72	pass	Memory usage within limits

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application successfully predicts loan approval based on applicant details. The prediction results are generated quickly and the web application responds smoothly with accurate output.

Bottlenecks Identified:

No major performance issues were observed during testing. The application loads the trained model only once at startup, and predictions are generated efficiently.

Optimization Steps Taken:

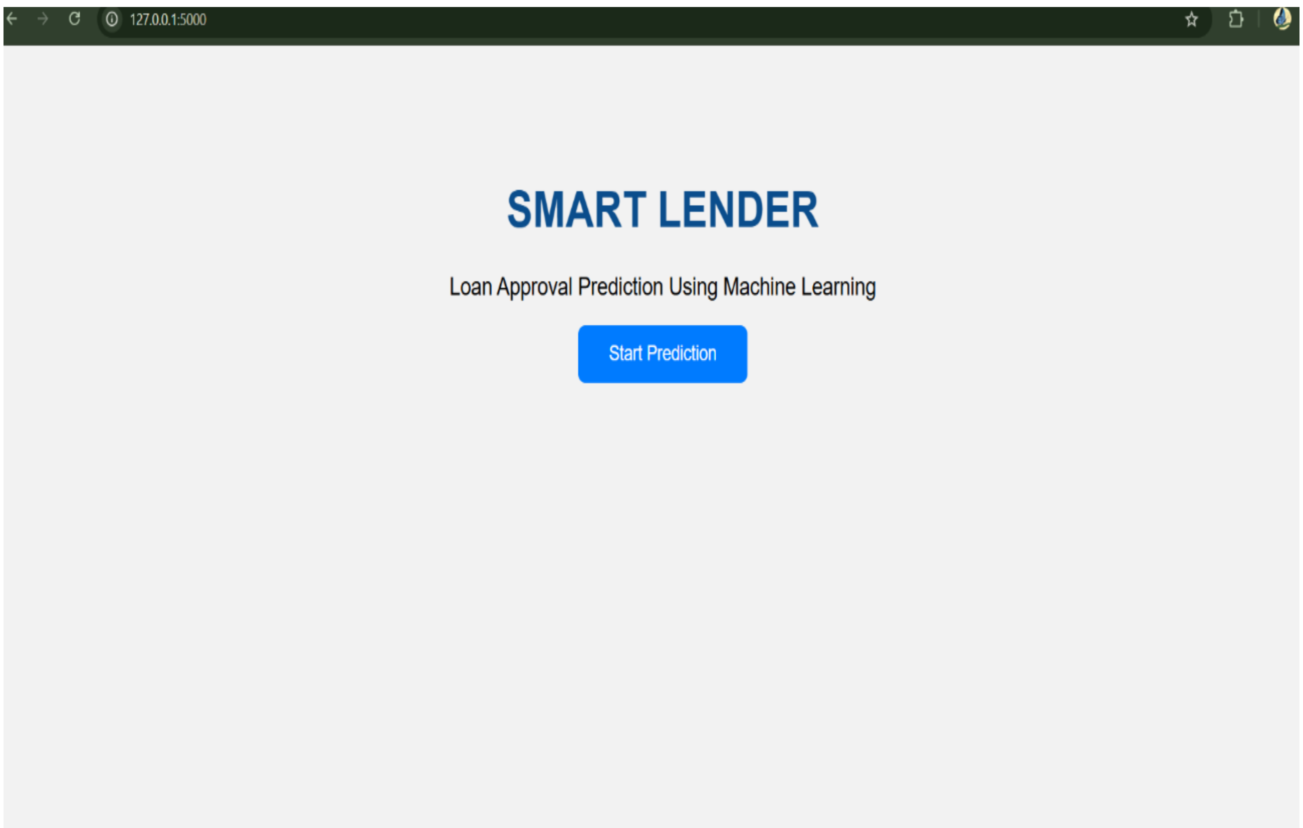
The trained Machine Learning model and StandardScaler were saved using Joblib to reduce prediction time. Efficient preprocessing and input validation were implemented to improve the overall performance of the application.

step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1:

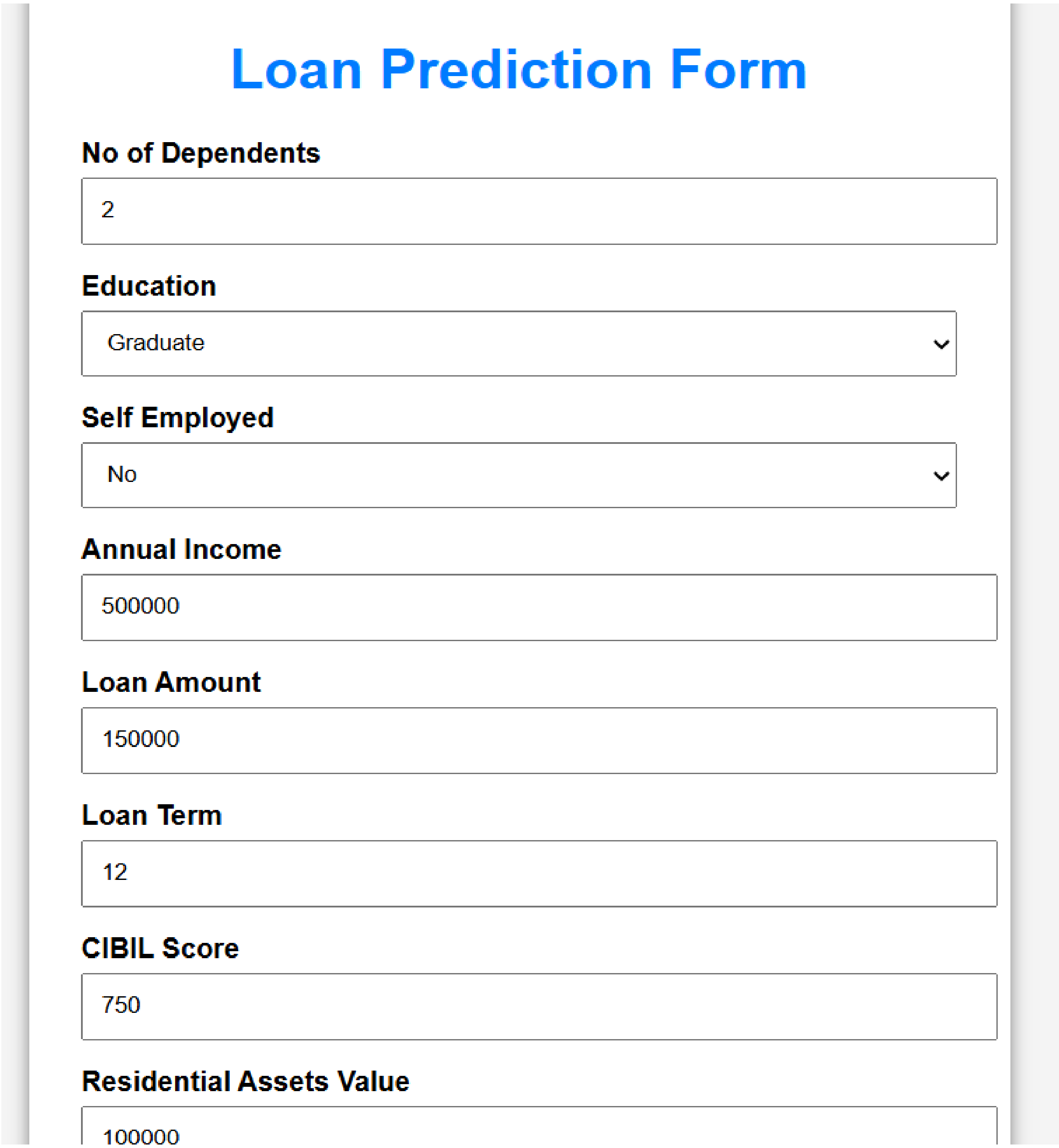
Smart Lender – Home Page: The home page of the Smart Lender application displaying the project title and the Start Prediction button.



Screenshot 2:

Smart Lender – Loan Prediction Input Form

The input form where users enter applicant details such as income, loan amount, CIBIL score, education, employment status, and other required information for loan prediction.



Screenshot 3

Figure 3: Smart Lender – Loan Approval Prediction Result

The prediction result page displaying the loan approval status along with the applicant's key details generated by the machine learning model.